

SUKIRTHAN CHANDRAKUMAR

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EXECUTIVE SUMMARY

Motivated IT undergraduate specializing in Artificial Intelligence with strong knowledge of Python, Machine Learning, NLP, Generative AI, and backend API development. Experienced in building AI-powered applications using LLMs, RAG, OpenAI APIs, Pinecone, Flask, Spring Boot, and MongoDB. Skilled in data preprocessing, model training, model evaluation, and developing full-stack AI solutions. Seeking an AI/ML Engineering Internship to apply machine learning knowledge, contribute to real-world AI projects, and grow in a collaborative technical environment.

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, HTML, CSS, SQL (Oracle, MySQL)
- **AI & Machine Learning:** Supervised & Unsupervised Learning, Linear Regression, Decision Trees, Support Vector Machines (SVM), Random Forest
- **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, RAG, Flask MERN Stack, Tailwind CSS, Spring Boot
- **Data & AI Concepts:** Data Cleaning & Preprocessing, Dataset Handling (CSV, JSON), Natural Language Processing (NLP), Generative AI & Large Language Models (LLMs)
- **Tools & Platforms:** Git, GitHub, Visual Studio Code (VS Code), Jupyter Notebook, Google Colab, Pinecone, OpenAI API
- **Core Competencies:** API-Driven Architecture, Design Patterns, Data Cleaning, Feature Engineering, Model Training, Model Evaluation, REST API Development, OOP, DSA, Problem Solving

EDUCATION

Sri Lanka Institute of Information Technology

- BSc (Hons) in Information Technology – Specializing in Artificial Intelligence
- 2nd Year Student | CGPA: 3.2 / 4.0

2024-Present

PROJECTS

Duinophile – Full-Stack AI-Powered Course Management Platform | Java, Spring Boot, MongoDB, Python, CodeT5, ML, REST API

- Developed Duinophile, a full-stack AI-powered Arduino learning platform that helps beginners learn Arduino programming, debug code, and get support for hardware-related questions.
- Integrated an AI chatbot powered by a fine-tuned CodeT5-small model trained on 2,750+ Arduino error examples to detect coding mistakes and generate corrected code suggestions.
- Implemented Spring Boot backend APIs, MongoDB database integration, and a Python-based ML server for real-time AI chatbot responses.

End-to-End Medical Chatbot GenAI | OpenAI, Pinecone, Flask, LLMs, RAG ,Python

- Developed an AI-powered medical chatbot using LLMs for intelligent question answering, integrating OpenAI for natural language understanding and Pinecone for efficient vector search.
- Built backend APIs with Flask to retrieve relevant medical data and generate accurate, context-aware responses, improving user interaction and response quality.

AI/ML Model Training with Data Preprocessing Pipeline | Python, Scikit-learn, XGBoost

- Developed an end-to-end ML pipeline including data preprocessing, feature engineering, model training, and evaluation on structured datasets.
- Implemented and compared models (Linear Regression, Decision Tree, Random Forest, SVR, XGBoost, MLP) using MAE and R² to select the best-performing model for accurate predictions.

AI Summarizer | Python, Flask, React, OpenAI APIs

- Developed a full-stack AI-powered summarization platform to generate concise summaries from plain text and PDF documents.
- Integrated OpenAI APIs and modern NLP techniques to extract key insights, with a React frontend and Flask backend for seamless user interaction.

Achievements

- [View my certificates here](#)

Languages

- English(Advanced)
- Tamil(Native)
- Sinhala(Basic)